CHENG ZHU

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EDUCATION

University of Pennsylvania, Philadelphia, PA — Sep. 2025 – May 2027 (Expected)
Master of Science in Engineering (M.S.E.) in Robotics

University of Illinois Urbana-Champaign, Champaign, IL — Sep. 2021 – Jun. 2025
Bachelor of Science in Computer Engineering | Graduated with Highest Honors; Dean's List

PUBLICATIONS & MANUSCRIPTS

1. Haonan Chen, **Cheng Zhu**, Shuijing Liu, Yunzhu Li, and Katherine Rose Driggs-Campbell. *"Tool-as-Interface: Learning Robot Policies from Observing Human Tool Use."* Proceedings of the 9th Conference on Robot Learning (CoRL), PMLR 305:2695–2713, 2025.
   Award: Best Paper Award, ICRA 2025 Workshop on Foundation Models and Neuro-Symbolic AI for Robotics.
2. **Cheng Zhu**, Haonan Chen, Nadia Figueroa*, and Yilun Du*. *"Dynamical-System Policy for Robot Manipulation."*arXiv preprint forthcoming.
   ** Equal advising.*

RESEARCH EXPERIENCE

Dynamical-System Policy for Robot Manipulation — May 2026 – Present
Graduate Researcher | Co-Advisors: Prof. Nadia Figueroa and Prof. Yilun Du — ***Figueroa Robotics Lab, UPenn*** ***Embodied Minds Lab, Harvard***

- Developed a dynamical-system-based policy for closed-loop motion generation in robotic manipulation.
- Implemented the training, simulation, and evaluation pipeline for dynamical-system action generation.

Dynamical-System Action Tokenization for Vision-Language-Action Models — 2026 – Present
Graduate Researcher | Advisor: Prof. Nadia Figueroa — ***Figueroa Robotics Lab, UPenn***

- Designed a goal-centric SEDS tokenizer that converts long-horizon robot demonstrations into compact, executable dynamical-system tokens.
- Built a pipeline for skill parsing, phase segmentation, trajectory clustering, SEDS fitting, and parameter quantization across left-arm, right-arm, and mobile-base streams.
- Evaluated decoded motion primitives on Behavior-1K demonstrations using post-quantization convergence tests and simulation rollouts.

Composable Action Representations with Koopman Operators — Jan. 2026 – May 2026
Graduate Researcher | Advisor: Prof. Nadia Figueroa — ***Figueroa Robotics Lab, UPenn***

- Investigated learned Koopman operators as latent action representations for composing robot motion primitives.
- Implemented an encode–evolve–decode framework to evaluate action reconstruction and composition in the learned latent space.
- Identified operator approximation error and information loss during decoding as key bottlenecks to reliable action composition.

Tool-as-Interface: Learning Robot Policies from Observing Human Tool Use — Aug. 2024 – May 2025
Senior Researcher & Thesis Student | Advisor: Prof. Katherine Driggs-Campbell — ***Human-Centered Autonomy Lab, UIUC***

- Co-developed a framework that transfers human tool-use demonstrations to robot policies through 3D scene reconstruction, viewpoint augmentation, segmented observations, and tool-centric task-space actions.
- Integrated Grounded-SAM, SpatialTracker, and ViewCrafter; trained a vision transformer and explored RGB-depth fusion for robust cross-view representations.
- Implemented UR5e and Kinova robot interfaces and designed 3D-printed components for real-world experiments.
- Achieved a 71% improvement in average task success and reduced data-collection time by 77% versus time-matched teleoperation baselines.

Ringbot Motion Control — Aug. 2024 – Dec. 2024
Independent Study Researcher | Advisor: Prof. Joohyung Kim — ***KIMLAB (Kinetic Intelligent Machine Lab), UIUC***

- Improved decoupled speed and steering controllers and extended the finite-state machine for self-righting and fall-recovery behaviors.
- Implemented robot control software and supported hardware integration, wiring, and soldering.

RESEARCH EXPERIENCE (CONTINUED)

MiniHawk UAV Path Planning — May 2024 – Aug. 2024
Undergraduate Researcher | Advisor: Prof. Naira Hovakimyan — ***Advanced Controls Research Lab, UIUC***

- Reproduced and implemented Reverse Glowworm Swarm Optimization for UAV path planning in three-dimensional dynamic environments.
- Investigated trajectory-planning strategies for a VTOL MiniHawk platform using Linux, ROS, JAX, and Gazebo.

TECHNICAL SKILLS

Programming Python, C++, C, MATLAB, Java, R, SystemVerilog

Robotics & ML	ROS, Gazebo, JAX, SLAM, dynamical systems, Koopman operators, diffusion models, vision transformers
Platforms & Tools	UR5e, Kinova, Behavior-1K, Grounded-SAM, SpatialTracker, ViewCrafter, Linux
Hardware	3D printing, robot integration, wiring, soldering

Languages English, Chinese